Comptia N+

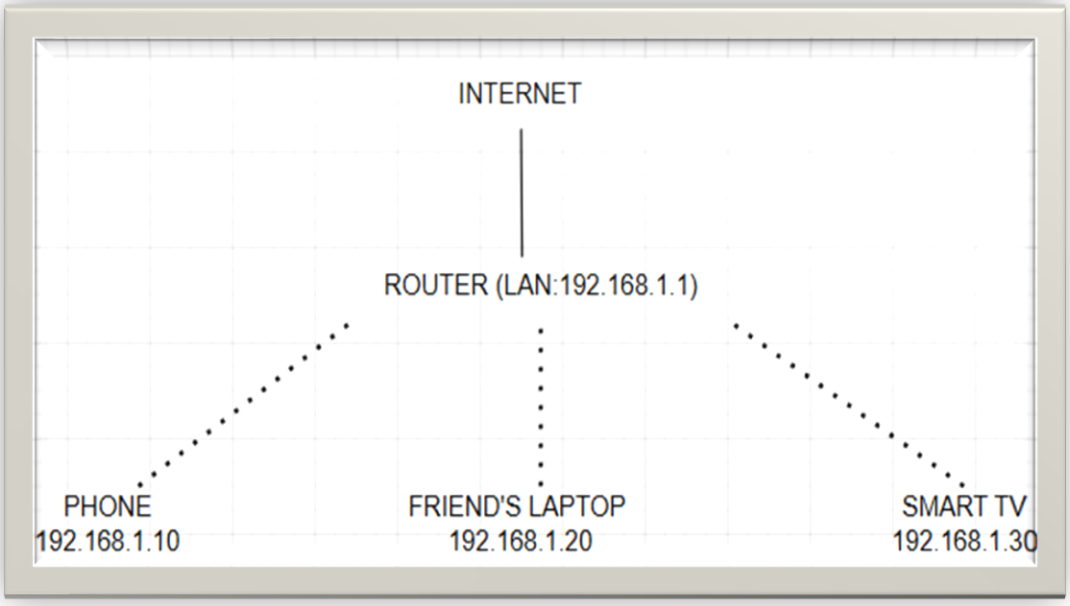
N+ NETWORK FUNDAMENTALS AND BUILDING NETWORKS

IP ROUTING AND IP PROTOCOLS

TASKS:1

Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.

RESULT



TASKS:2

Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made. Use made-up IP addresses and show the routing table entries that would be checked at each hop.

Comptia N+

RESULT

SCENARIO:

My laptop is on 192.168.1.10/24.

My friend's computer is on 192.168.2.20/24.

The router connects both networks.

STEPS:

1. My laptop creates a packet for 192.168.2.20.
2. The laptop checks its routing table.
3. Since the destination is on a different subnet, the packet is sent to the default gateway (192.168.1.1).
4. The router checks its routing table.
5. The router finds that the 192.168.2.0/24 network is connected through its second interface.
6. The router forwards the packet to the friend's computer.
7. The friend's computer receives the packet and sends a reply back through the same router.

Destination Network	Subnet Mask	Next Hop	Interface
192.168.1.0	255.255.255.0	Local	LAN1
192.168.2.0	255.255.255.0	Local	LAN2
0.0.0.0	0.0.0.0	ISP	WAN

Comptia N+

TASK3

Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

RESULT

Branch	Network Address	Subnet Mask	Next Hop	Interface
Delhi	10.0.1.0	255.255.255.0	Direct	LAN1
Mumbai	10.0.2.0	255.255.255.0	192.168.100.2	WAN
Indore	10.0.3.0	255.255.255.0	192.168.100.3	WAN

TASK4:

Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

RESULT

Static Routing	Dynamic Routing
Pros	
Easy to configure	Automatically updates routes
Uses less CPU and memory	Best for large networks

Comptia N+

More secure because routes do not change automatically	Quickly finds a new route if a link fails
Cons	
Must be updated manually	More complex to configure
Not suitable for large networks	Uses more CPU and memory
No automatic backup route	Can generate extra network traffic

EXAMPLE:

- **Static Routing:** Suitable for a **small office network**.
- **Dynamic Routing (OSPF/RIP):** Better for a **food delivery company like Zomato**, where many branches need automatic route updates.

TASKS:5

Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does. Paste the AI-generated config and annotate each part with a short description.

RESULT

SAMPLE CONFIGRATION:

Router(config)# router ospf 1

Comptia N+

Router(config-router)# network 192.168.1.0 0.0.0.255 area 0

Router(config-router)# network 192.168.2.0 0.0.0.255 area 0

Configuration Line	**Description**
router ospf 1	Starts the OSPF routing process.
network 192.168.1.0 0.0.0.255 area 0	Adds the 192.168.1.0 network to OSPF Area 0.
network 192.168.2.0 0.0.0.255 area 0	Adds the 192.168.2.0 network to OSPF Area 0.